

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

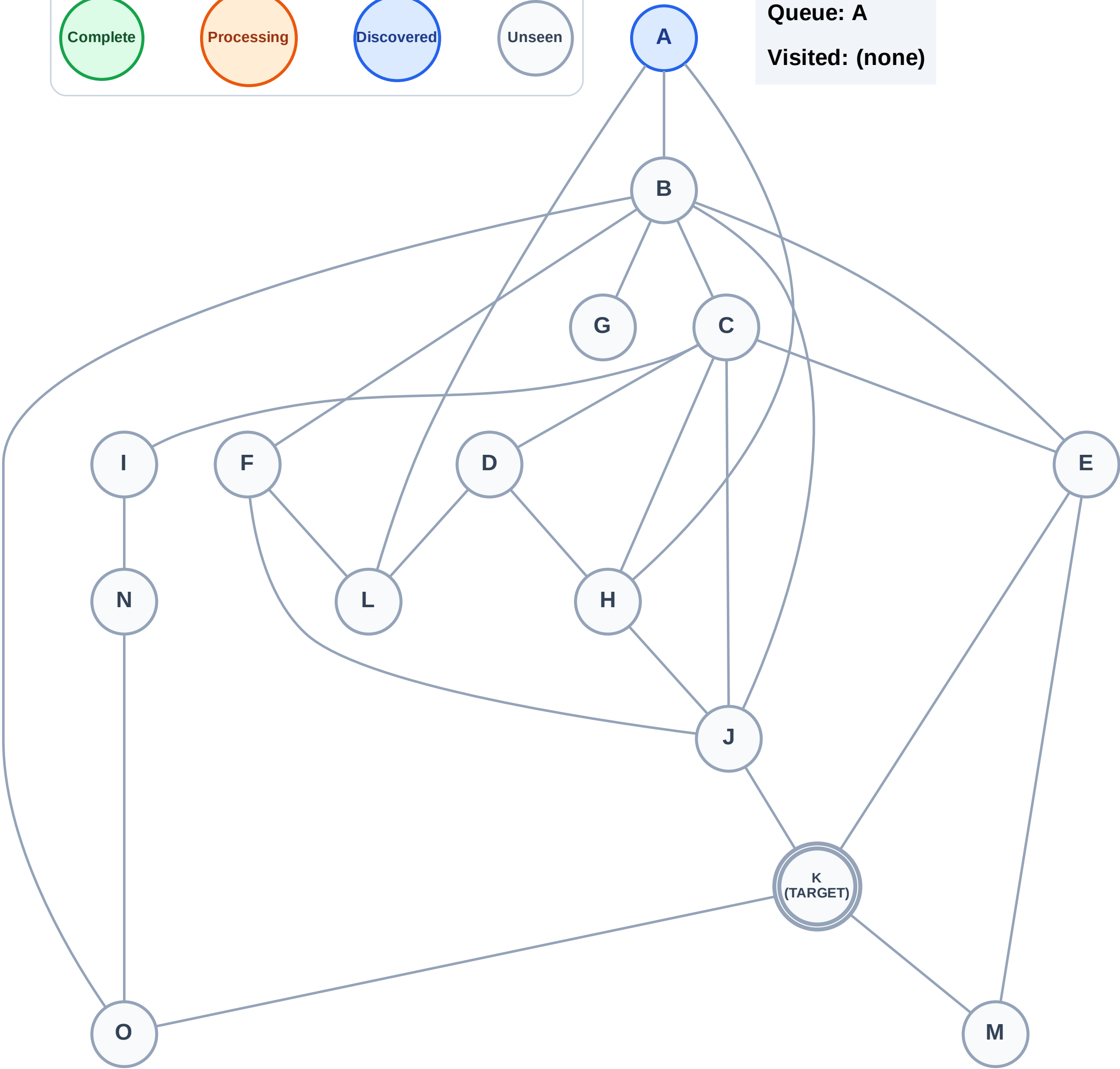
Complete

Processing

Discovered

Unseen

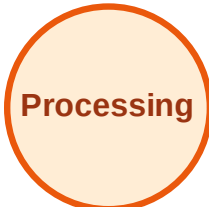
Queue: A
Visited: (none)



BFS Traversal

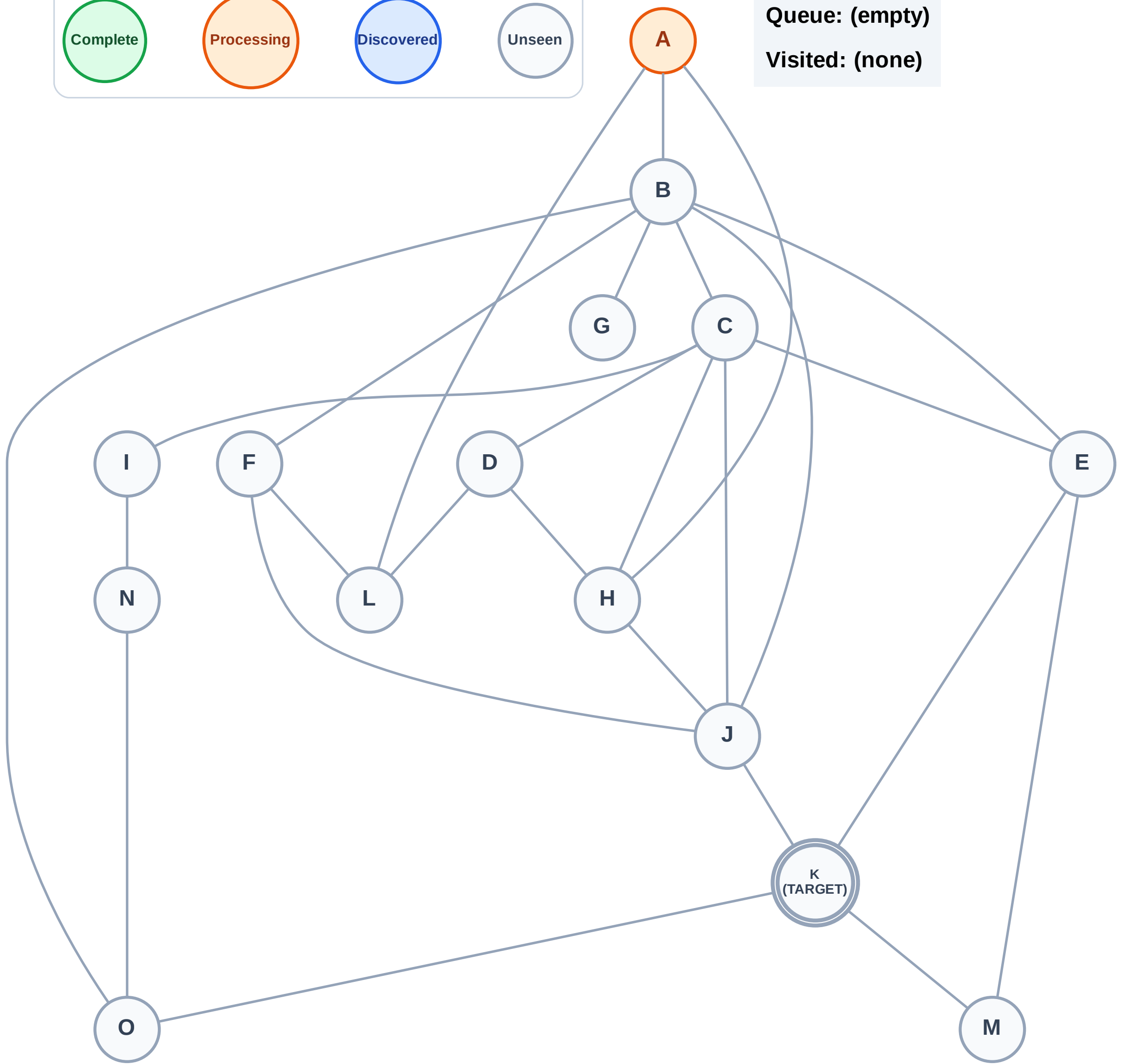
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, H, L from A

Legend

Complete

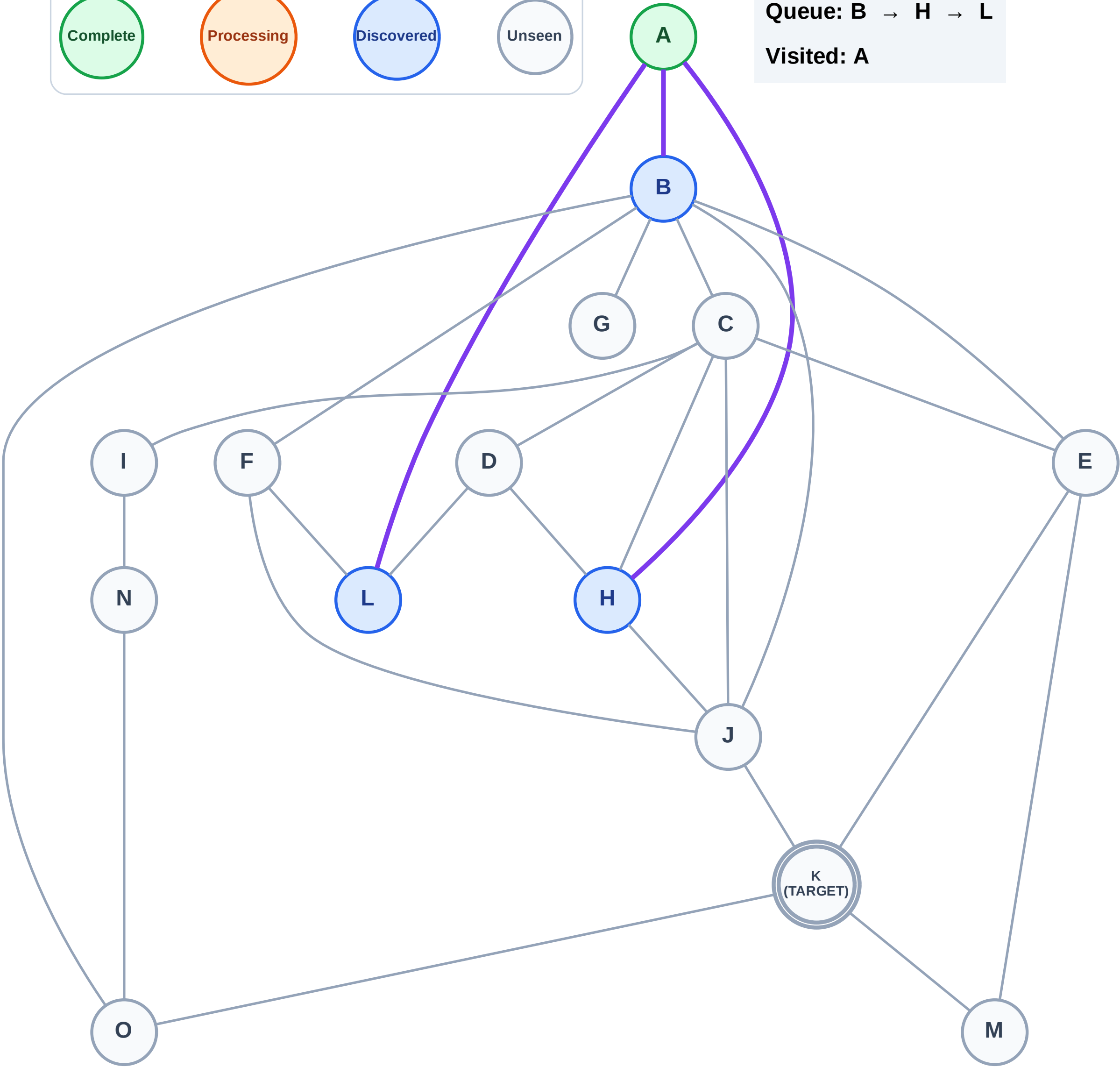
Processing

Discovered

Unseen

Queue: B → H → L

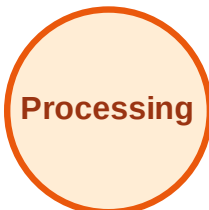
Visited: A



BFS Traversal

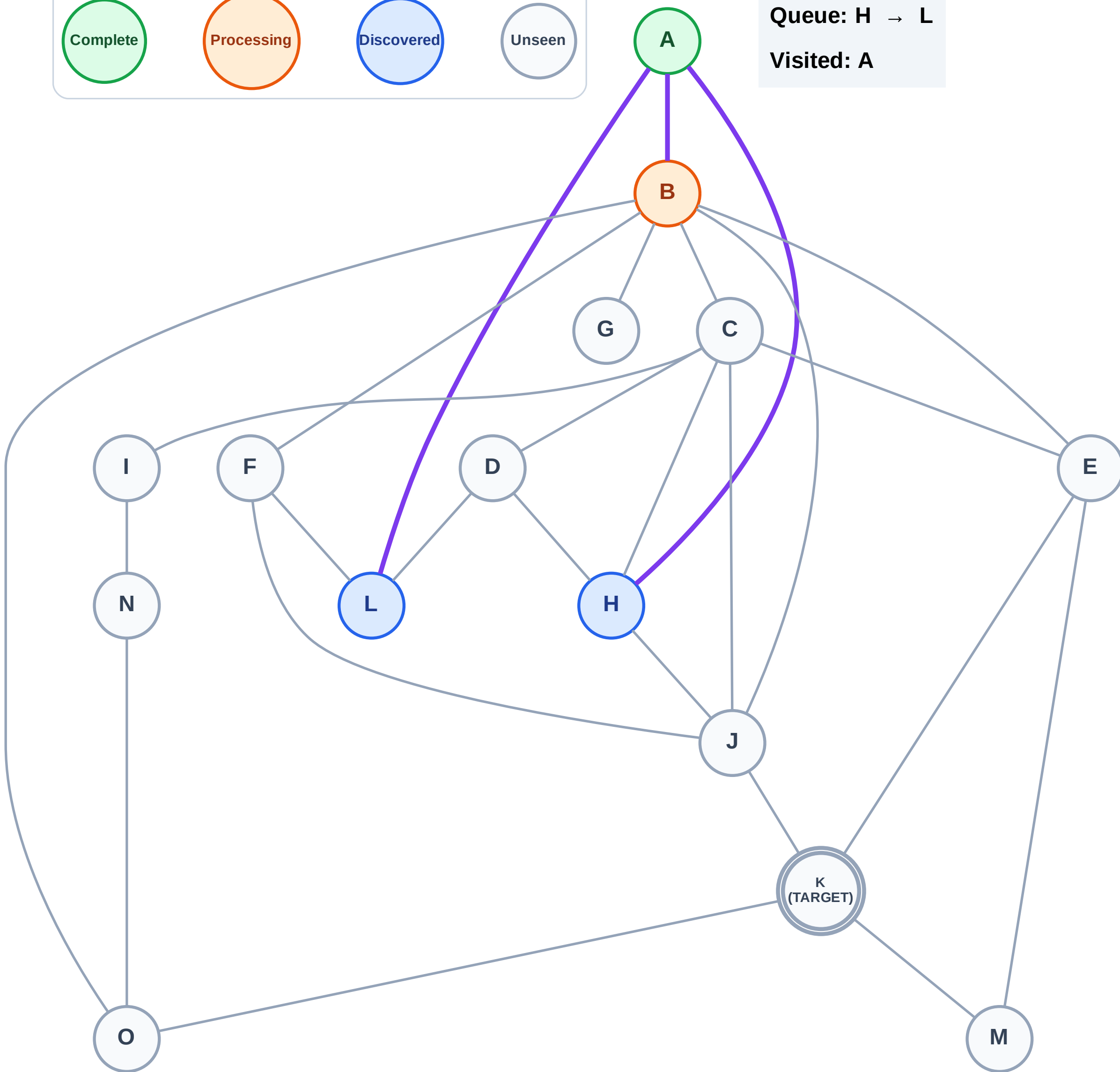
Step 4: Process node B

Legend



Queue: H → L

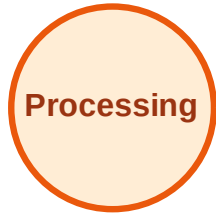
Visited: A



BFS Traversal

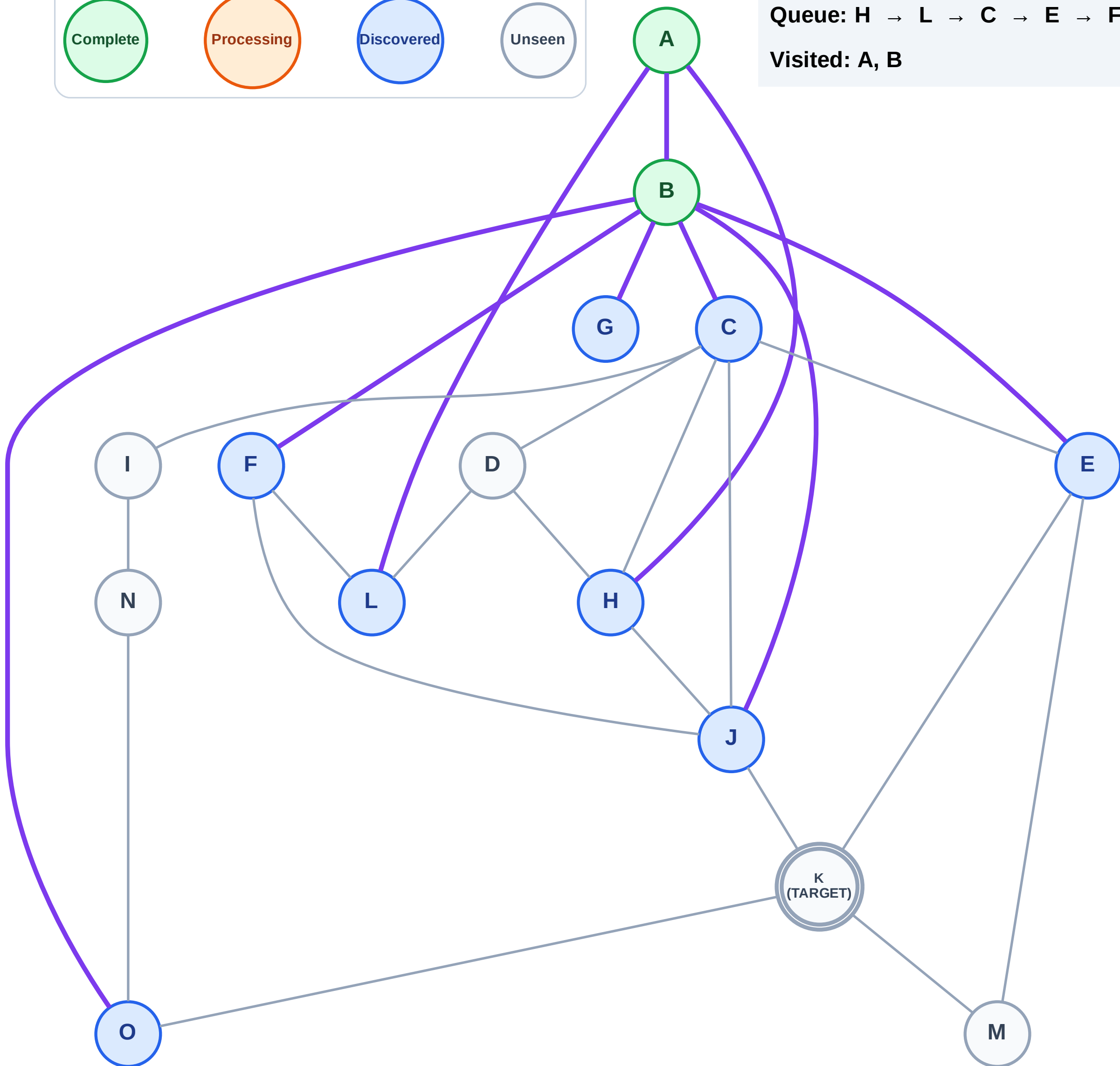
Step 5: Discover C, E, F, G, J, O from B

Legend



Queue: H → L → C → E → F → G → J → O

Visited: A, B



BFS Traversal

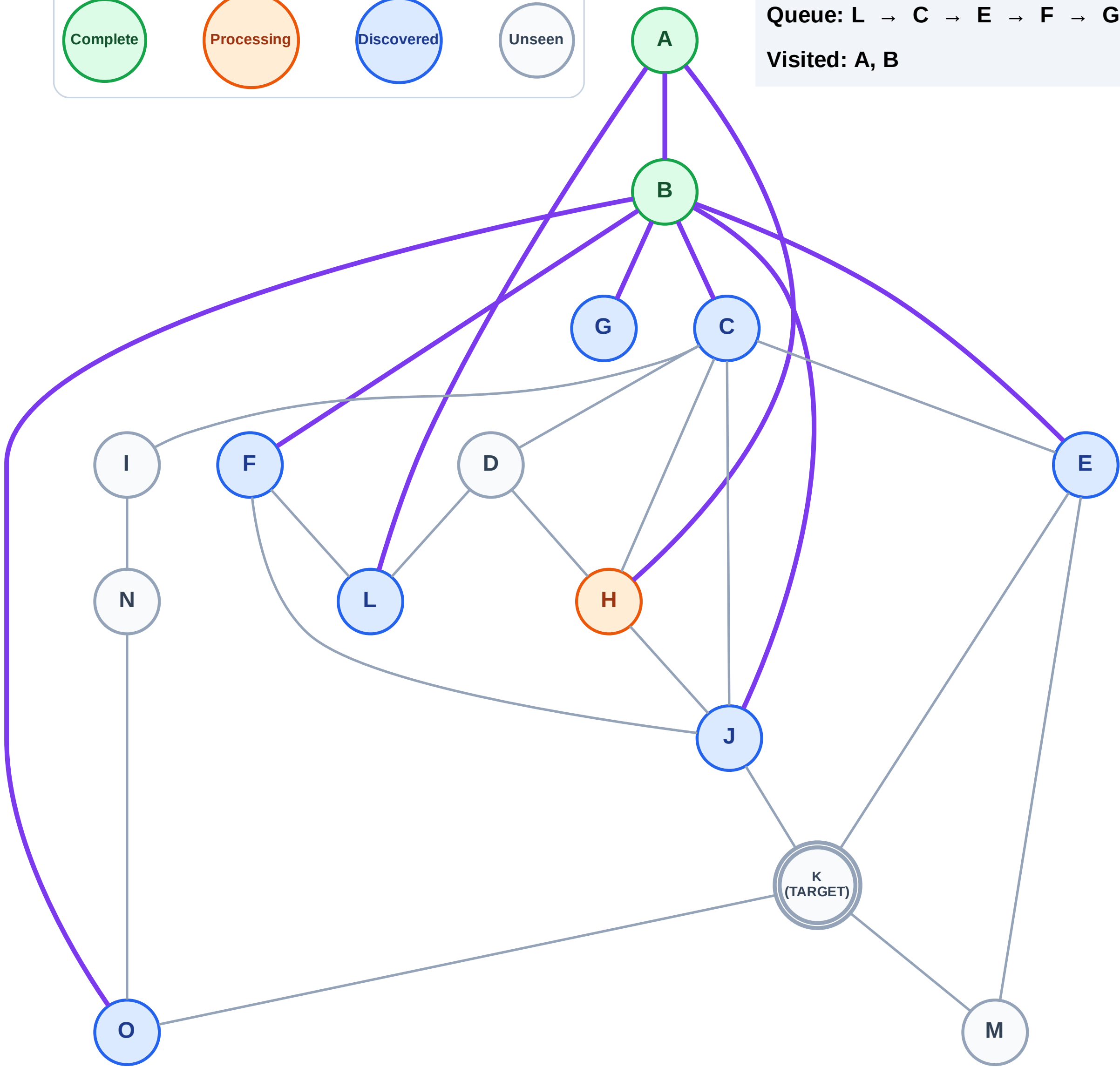
Step 6: Process node H

Legend



Queue: L → C → E → F → G → J → O

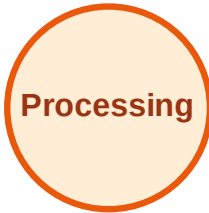
Visited: A, B



BFS Traversal

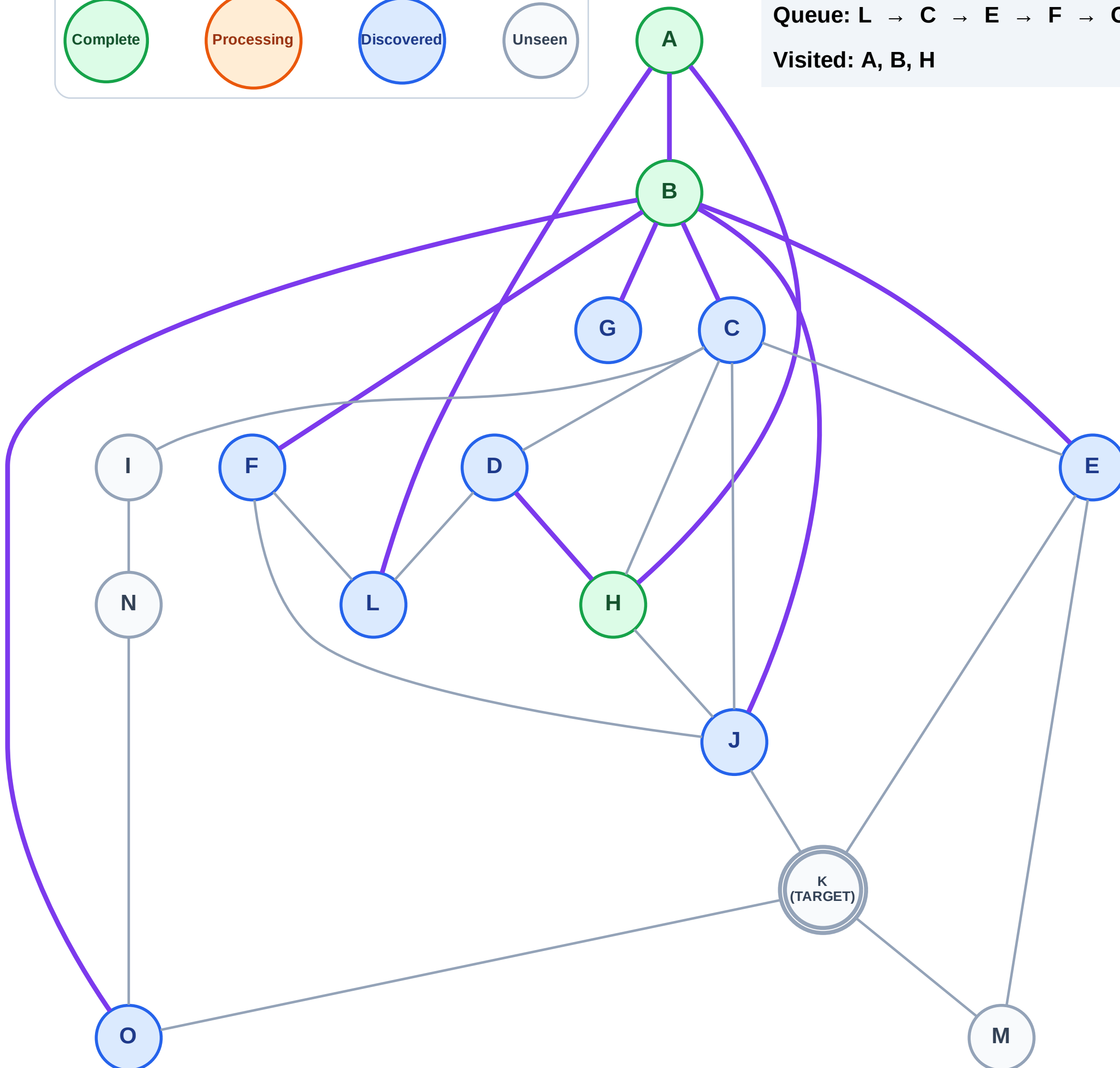
Step 7: Discover D from H

Legend



Queue: L → C → E → F → G → J → O → D

Visited: A, B, H



BFS Traversal

Step 8: Process node L

Legend

Complete

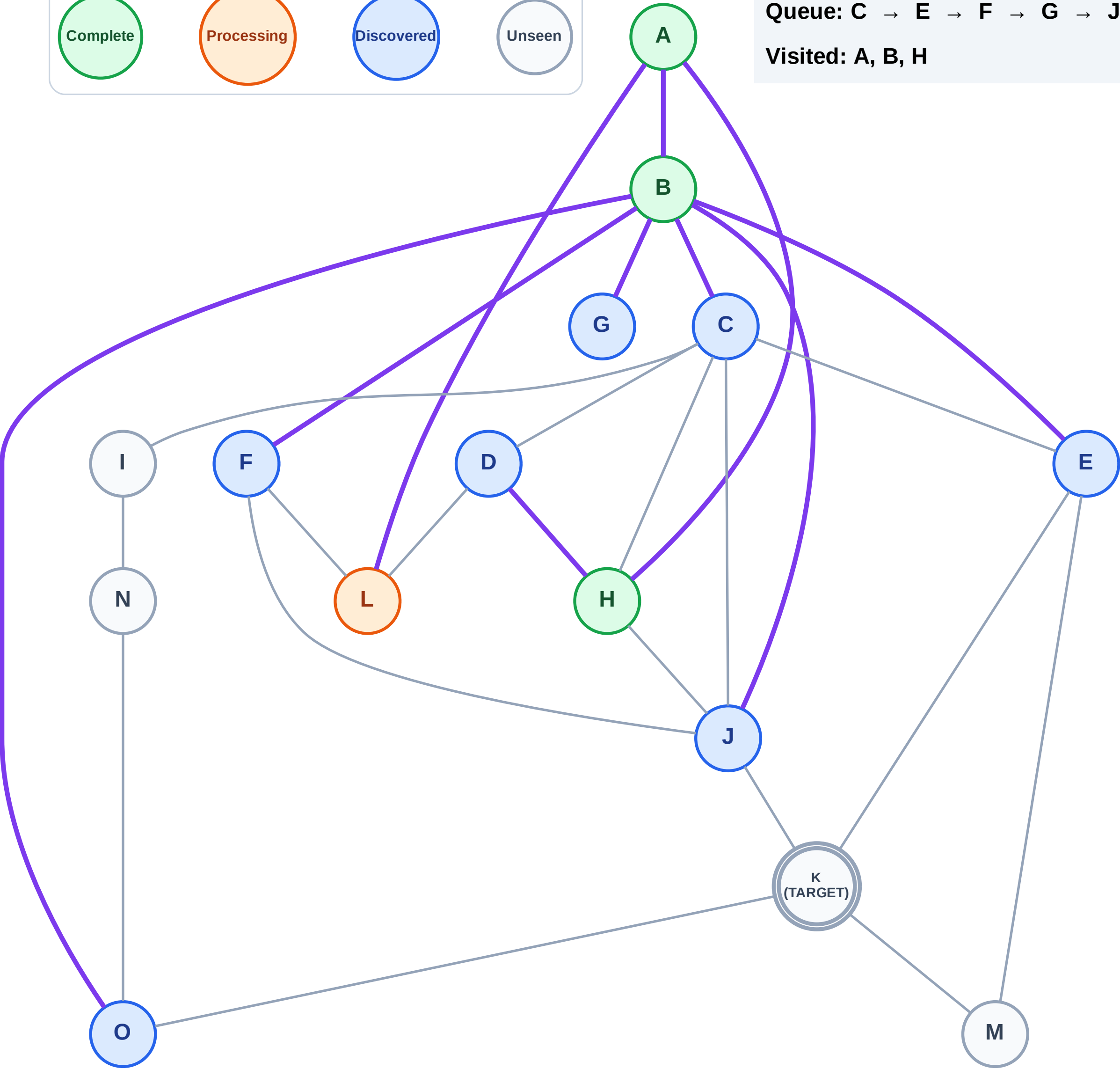
Processing

Discovered

Unseen

Queue: C → E → F → G → J → O → D

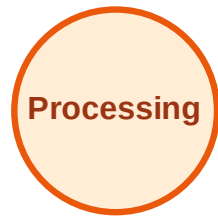
Visited: A, B, H



BFS Traversal

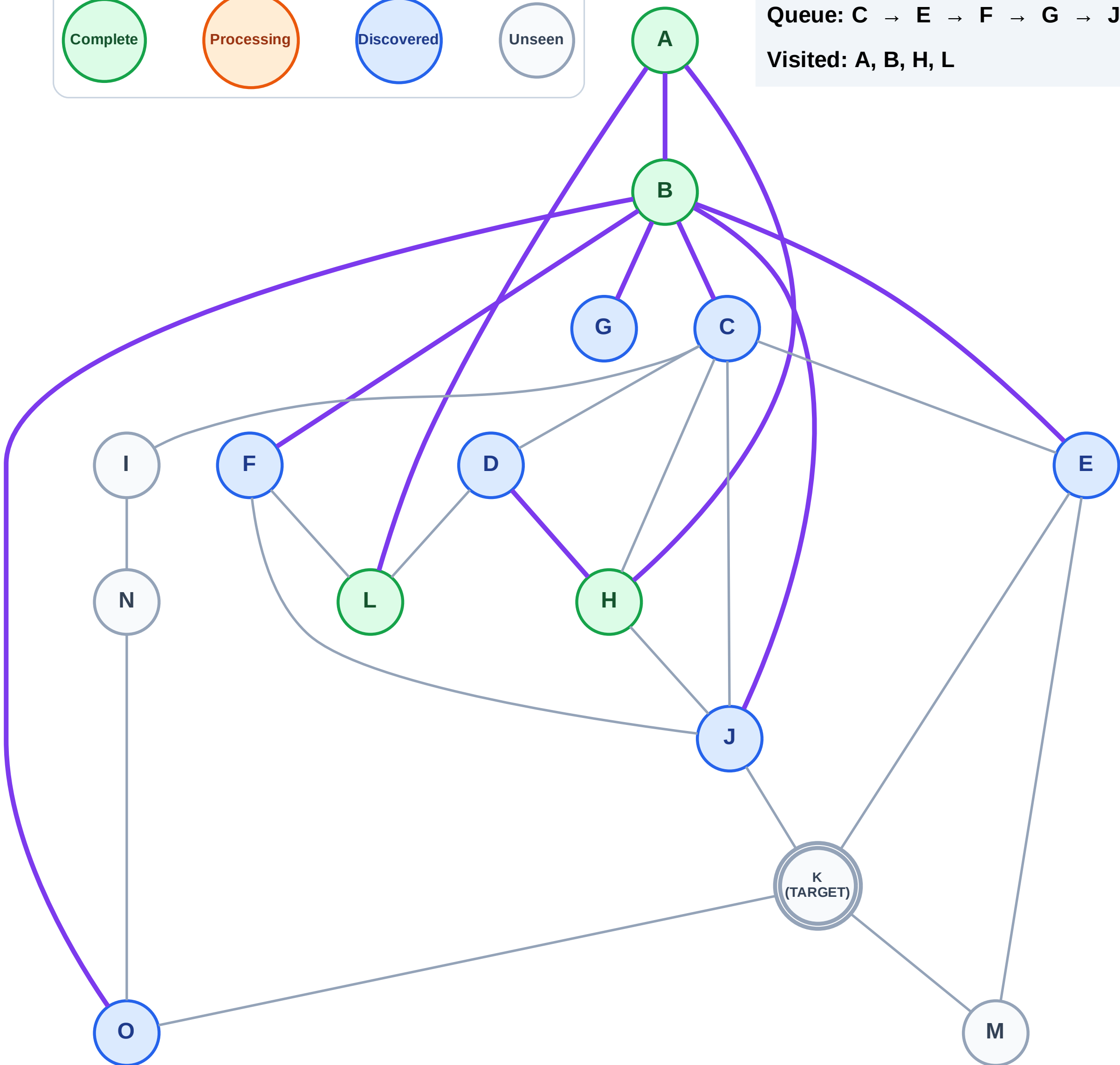
Step 9: No new neighbors from L

Legend



Queue: C → E → F → G → J → O → D

Visited: A, B, H, L



BFS Traversal

Step 10: Process node C

Legend

Complete

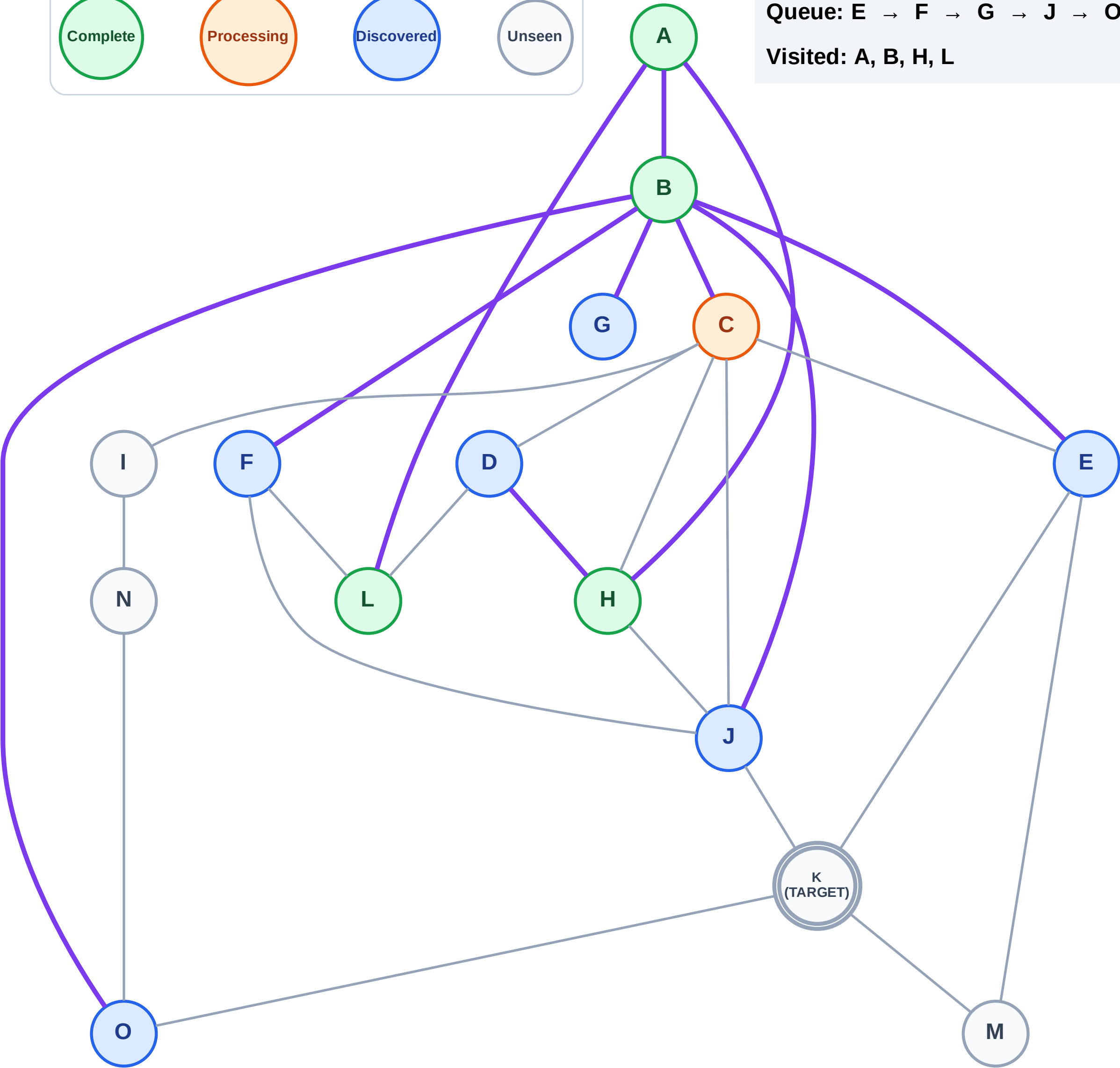
Processing

Discovered

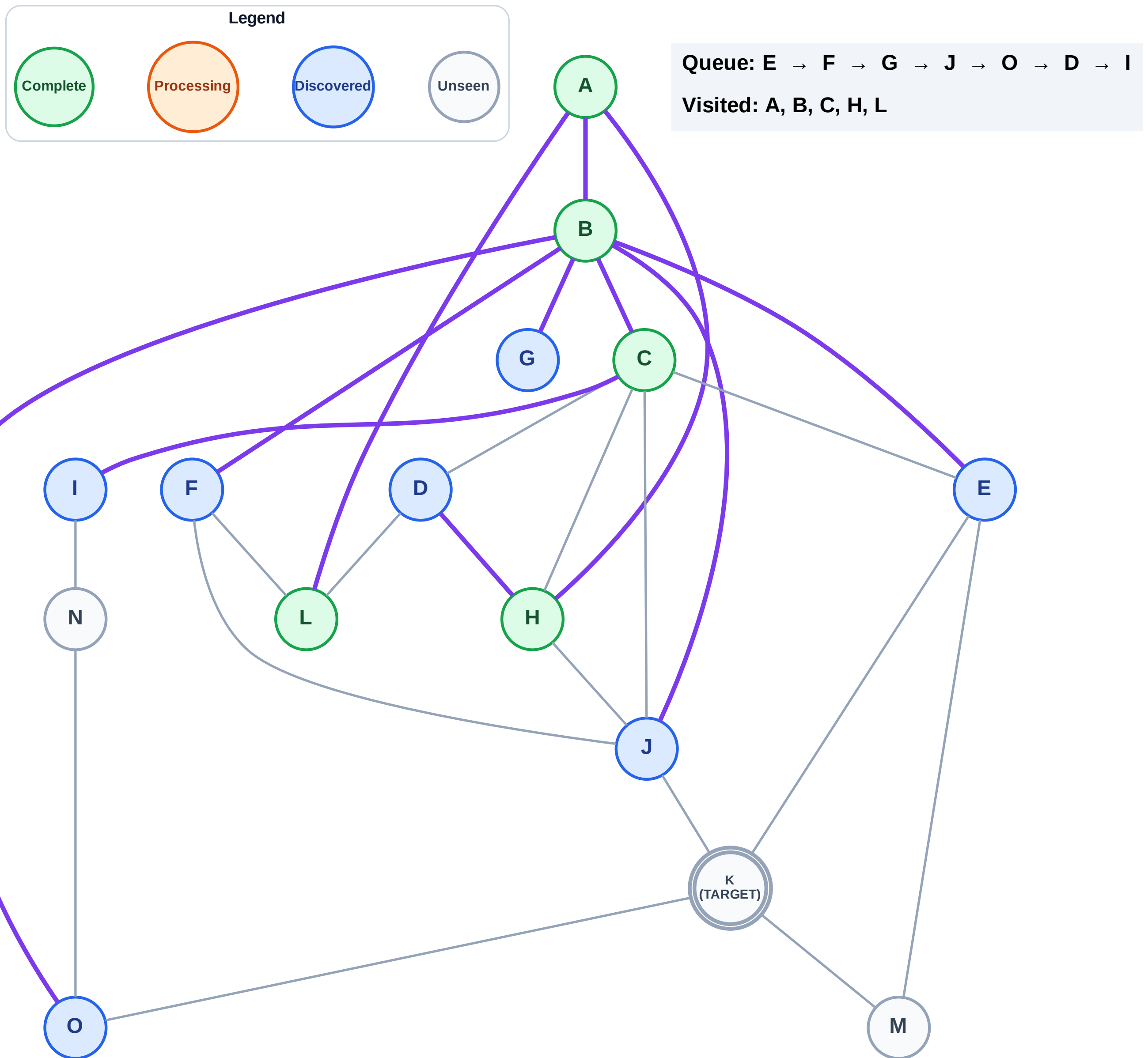
Unseen

Queue: E → F → G → J → O → D

Visited: A, B, H, L



Step 11: Discover I from C



BFS Traversal

Step 12: Process node E

Legend

Complete

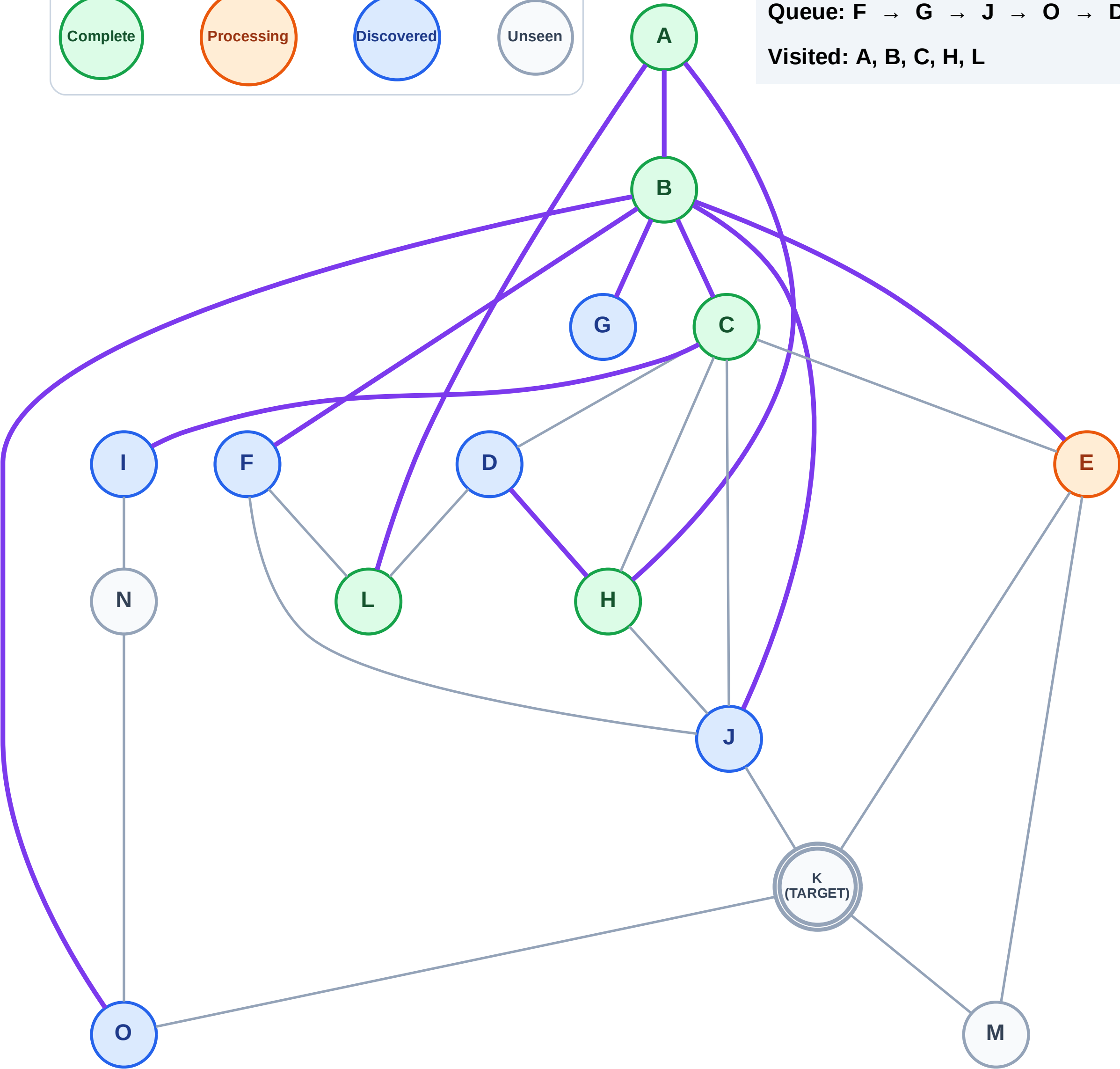
Processing

Discovered

Unseen

Queue: F → G → J → O → D → I

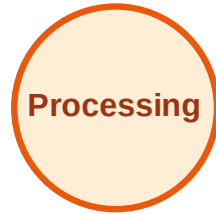
Visited: A, B, C, H, L



BFS Traversal

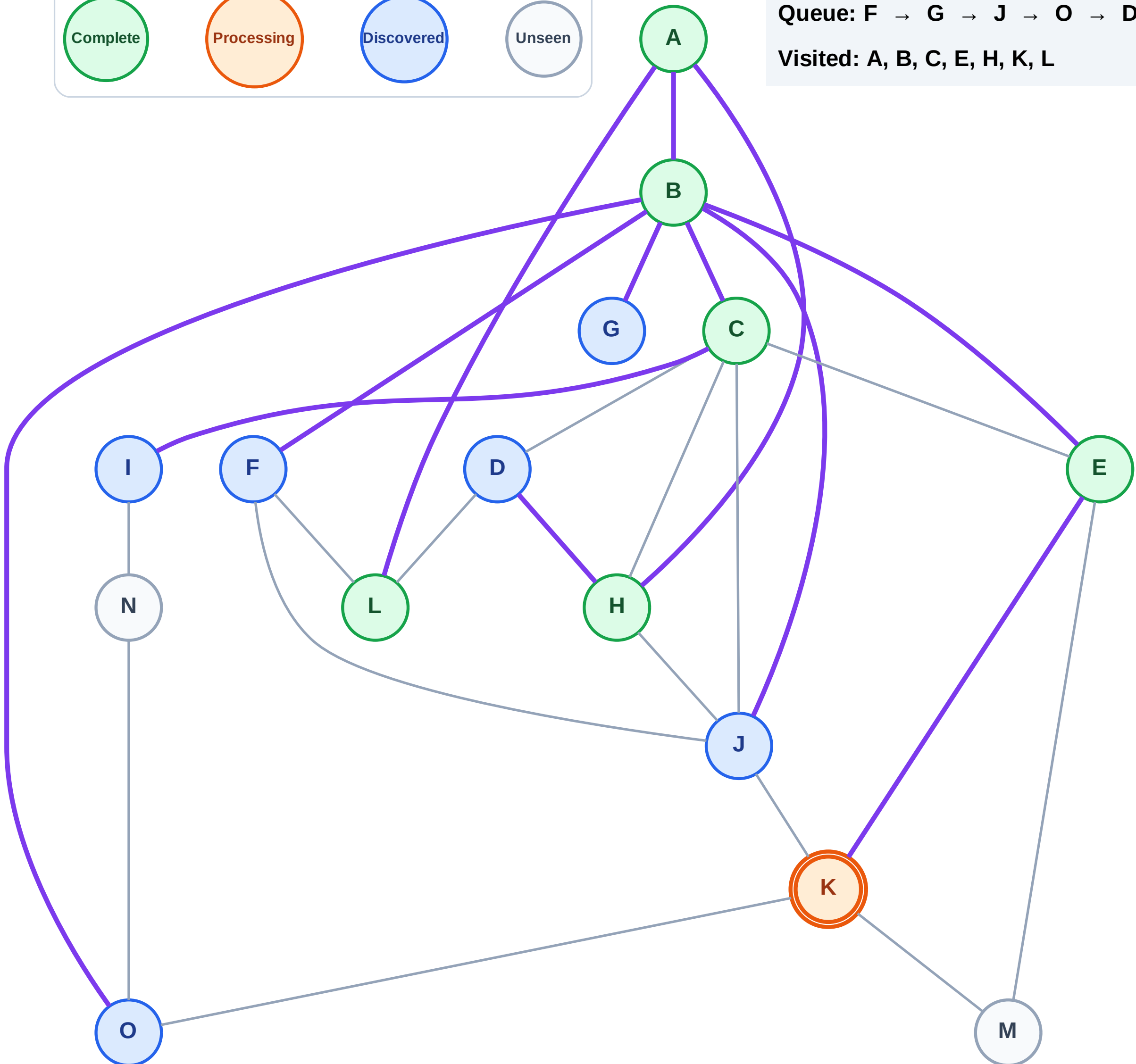
Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → E → K

Legend



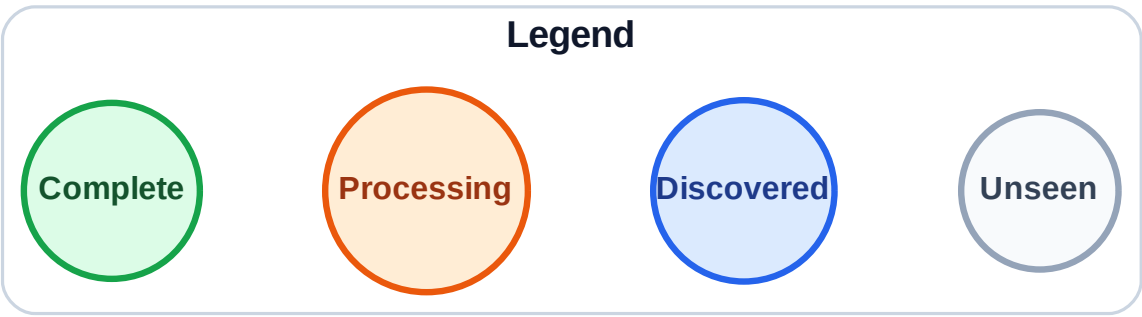
Queue: F → G → J → O → D → I

Visited: A, B, C, E, H, K, L

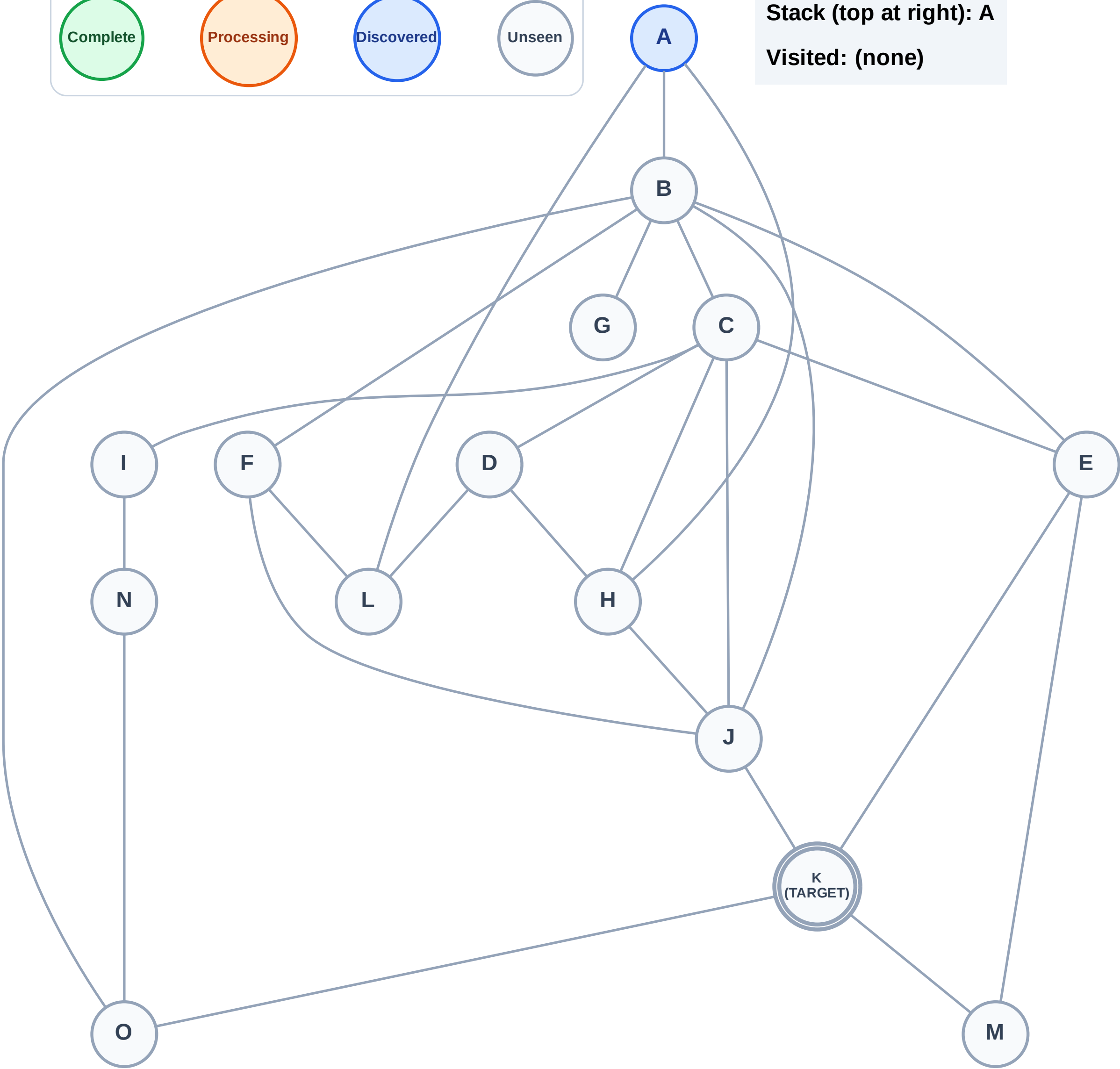


DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

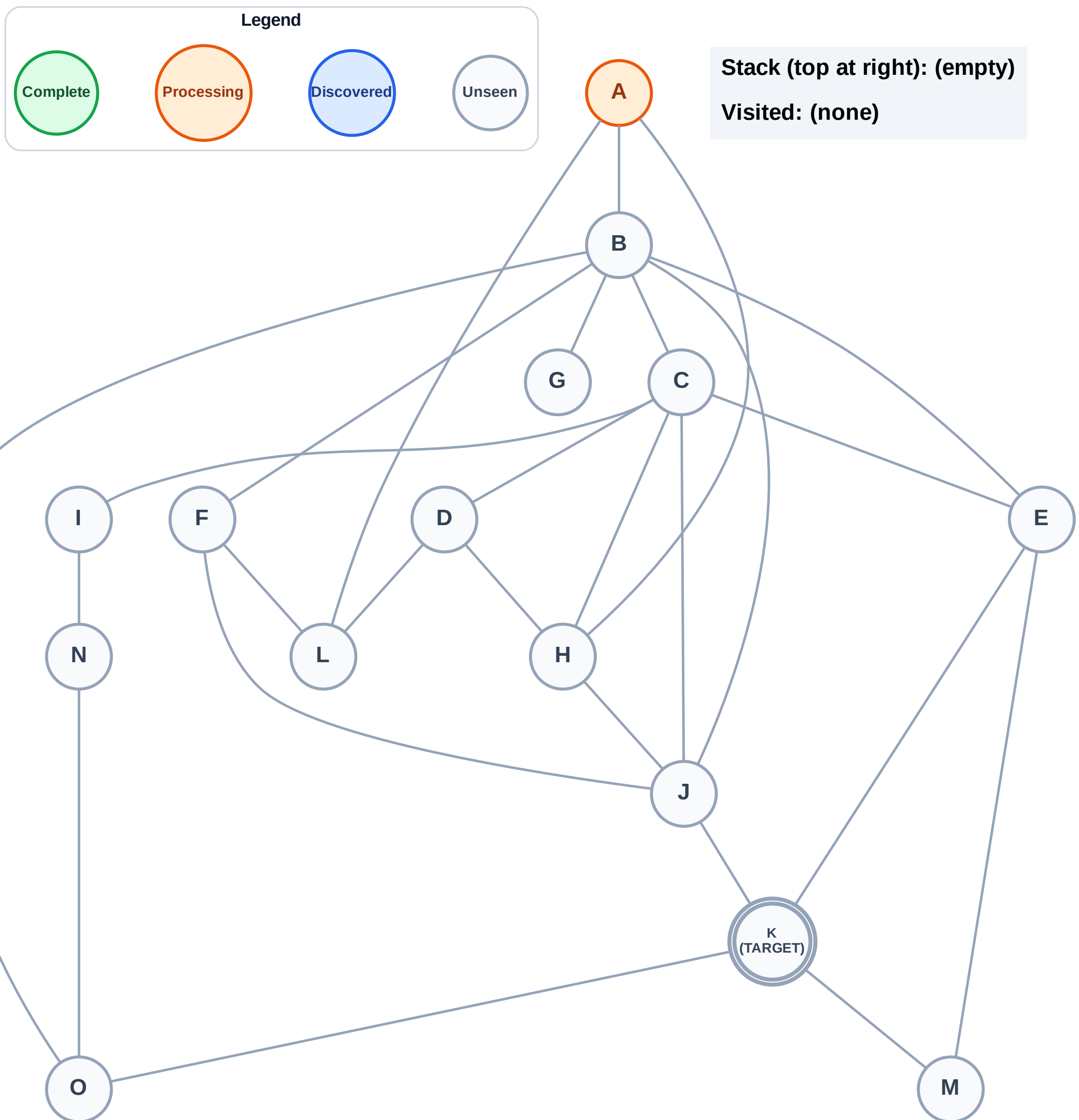


Stack (top at right): A
Visited: (none)



Step 2: Process node A

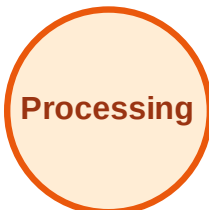
Step 2: Process node A



DFS Traversal

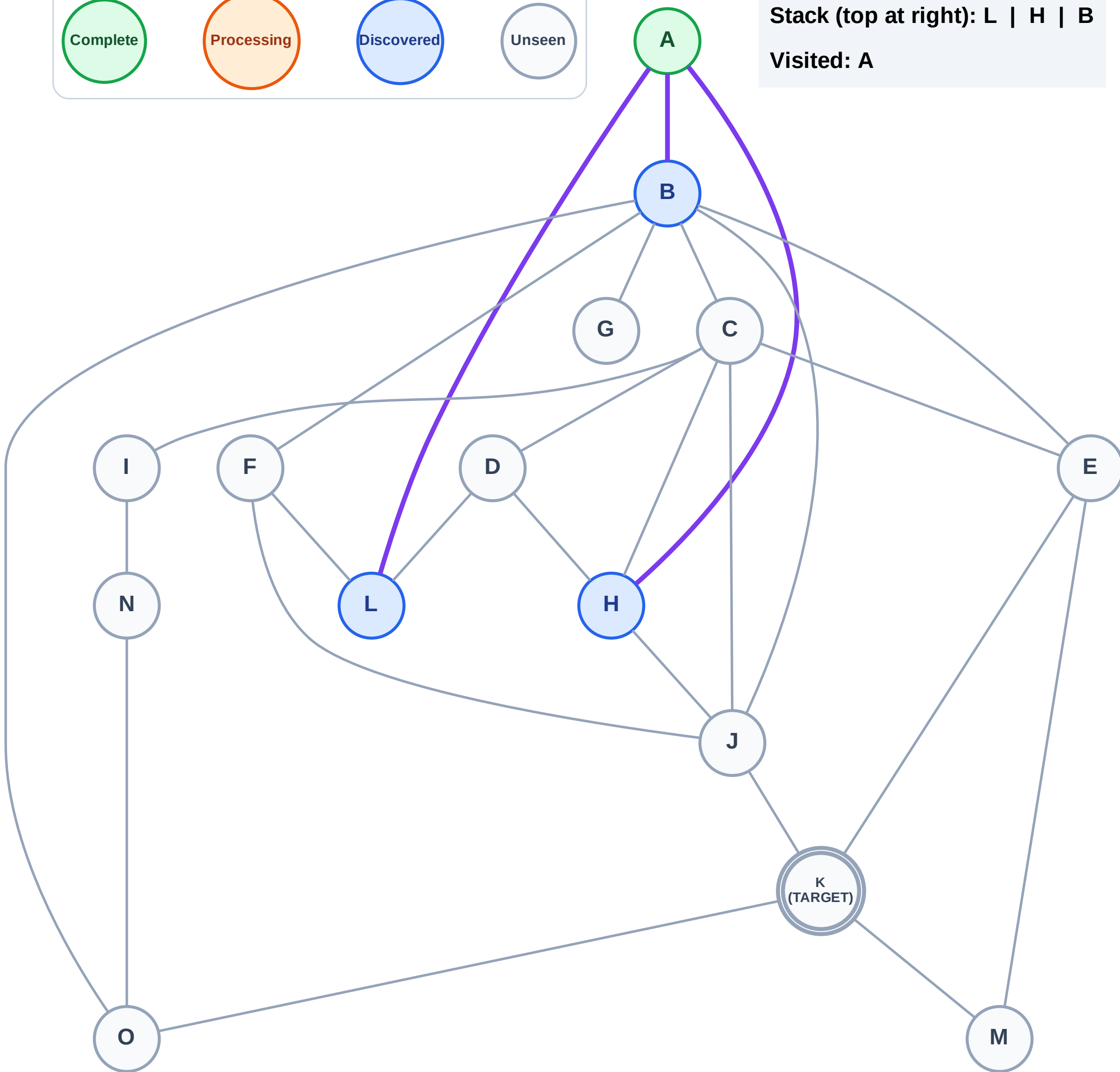
Step 3: Discover L, H, B from A

Legend



Stack (top at right): L | H | B

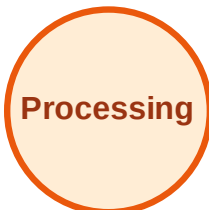
Visited: A



DFS Traversal

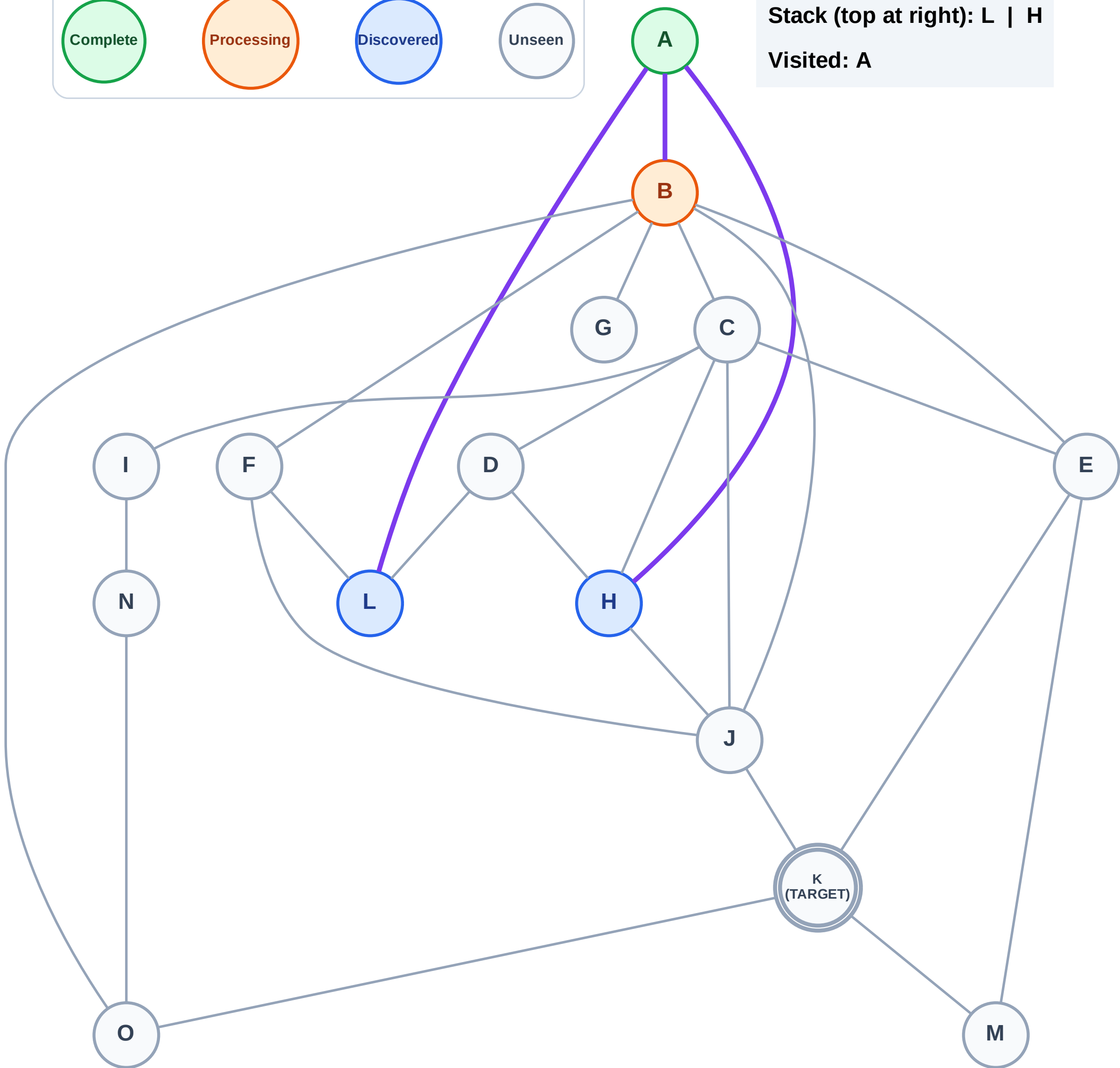
Step 4: Process node B

Legend



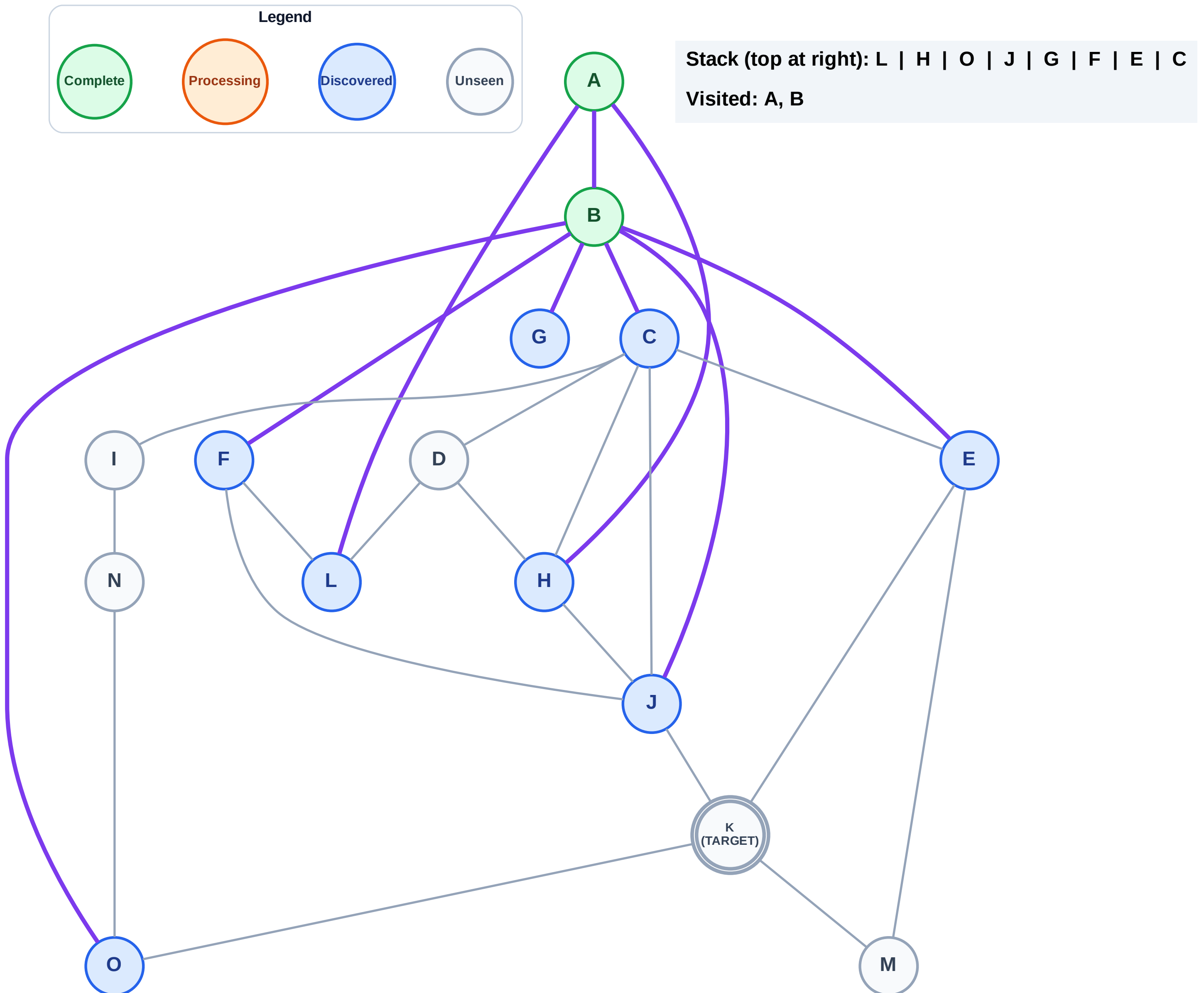
Stack (top at right): L | H

Visited: A



Step 5: Discover O, J, G, F, E, C from B

Step 5: Discover O, J, G, F, E, C from B



DFS Traversal

Step 6: Process node C

Legend

Complete

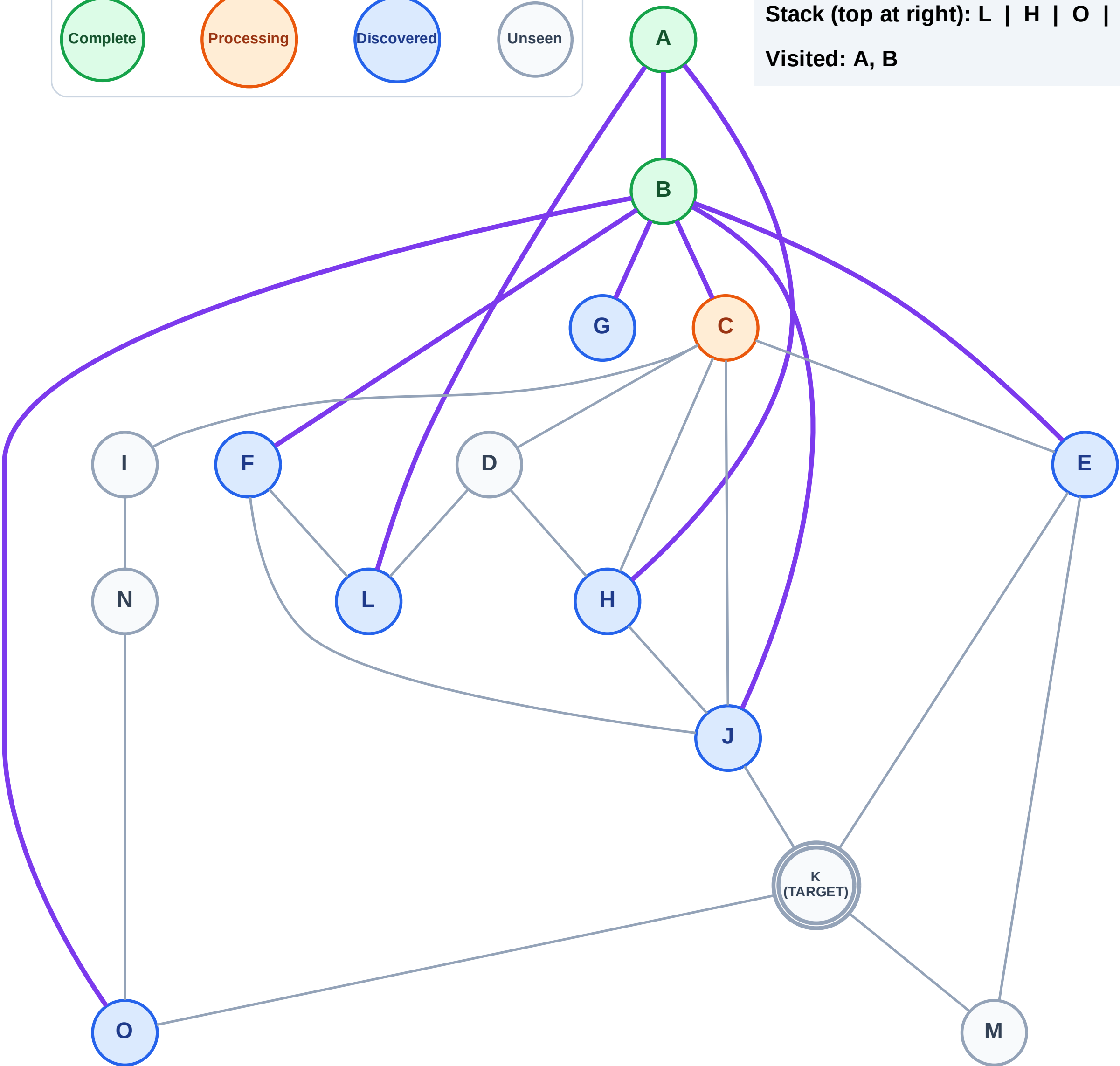
Processing

Discovered

Unseen

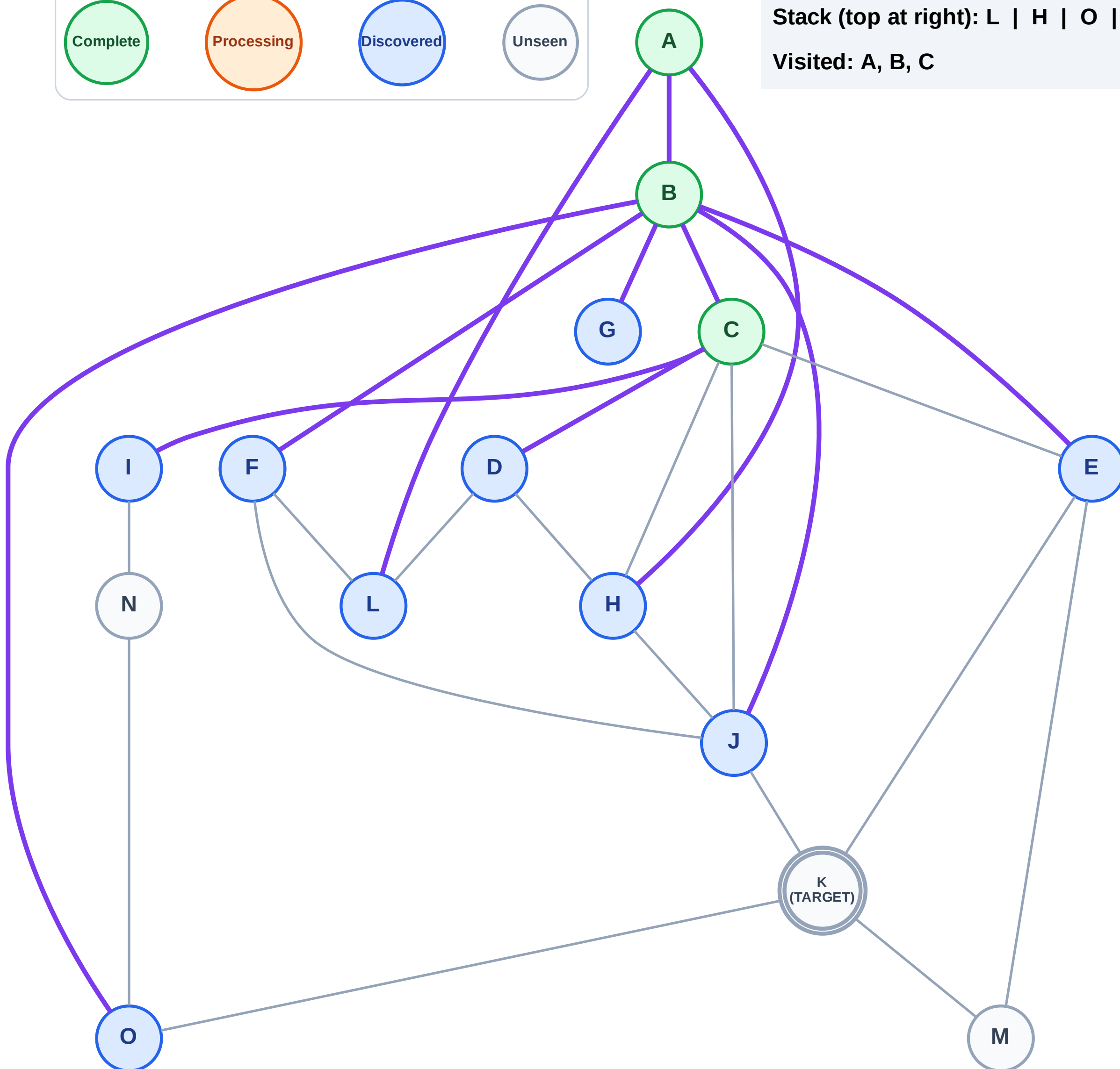
Stack (top at right): L | H | O | J | G | F | E

Visited: A, B



Step 7: Discover I, D from C

Step 7: Discover I, D from C



DFS Traversal

Step 8: Process node D

Legend

Complete

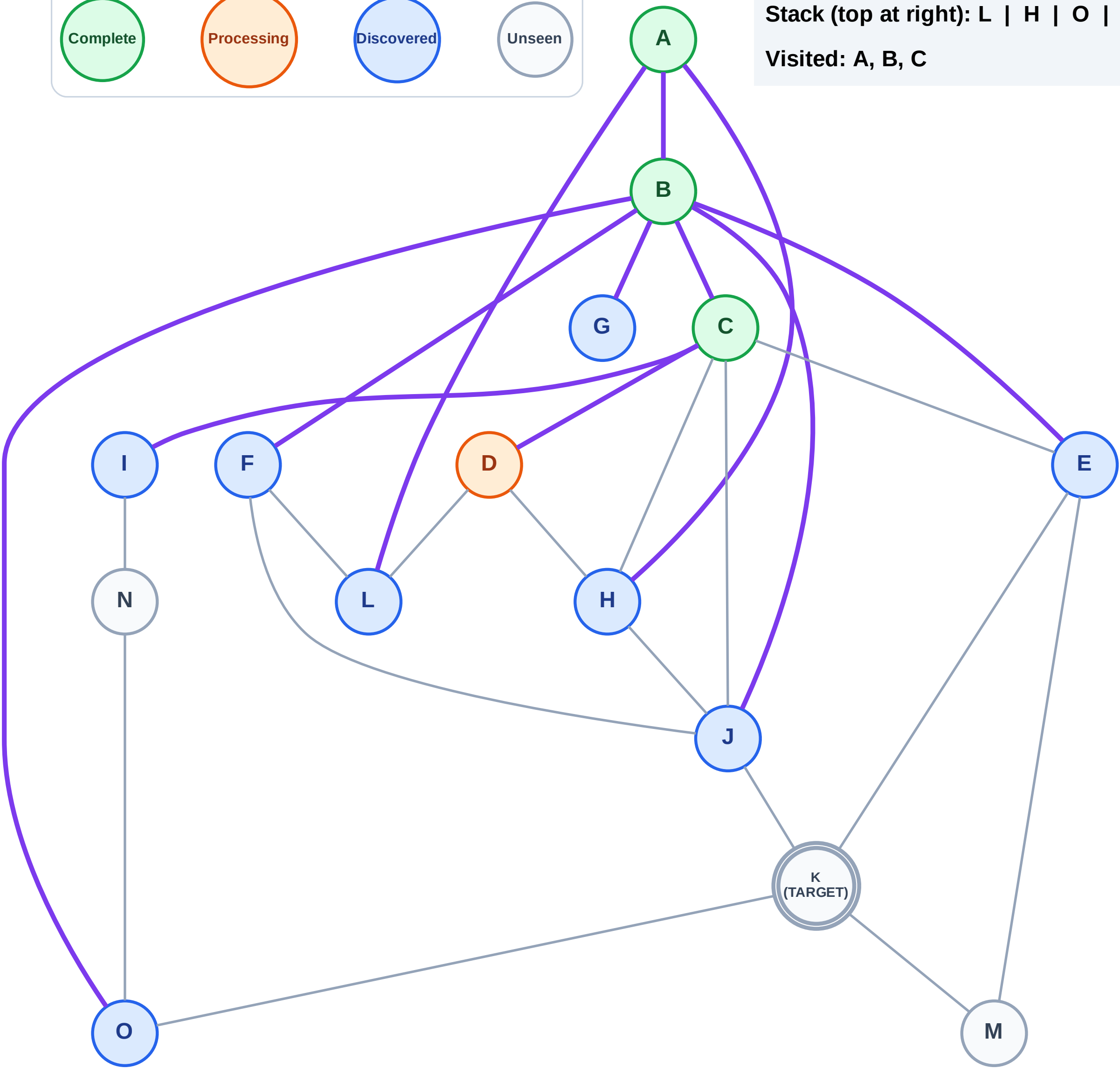
Processing

Discovered

Unseen

Stack (top at right): L | H | O | J | G | F | E | I

Visited: A, B, C



DFS Traversal

Step 9: No new neighbors from D

Legend

Complete

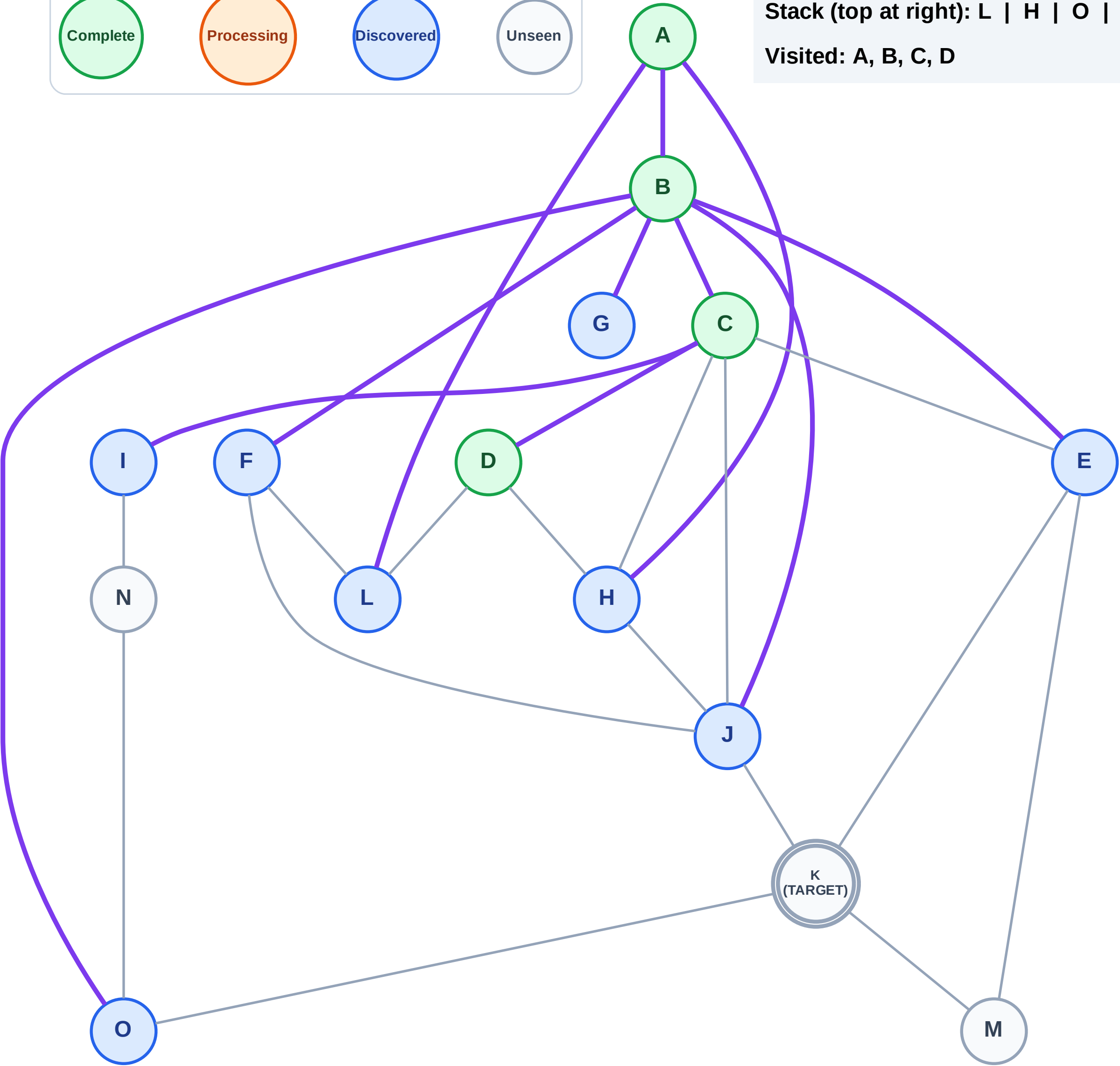
Processing

Discovered

Unseen

Stack (top at right): L | H | O | J | G | F | E | I

Visited: A, B, C, D



DFS Traversal

Step 10: Process node I

Legend

Complete

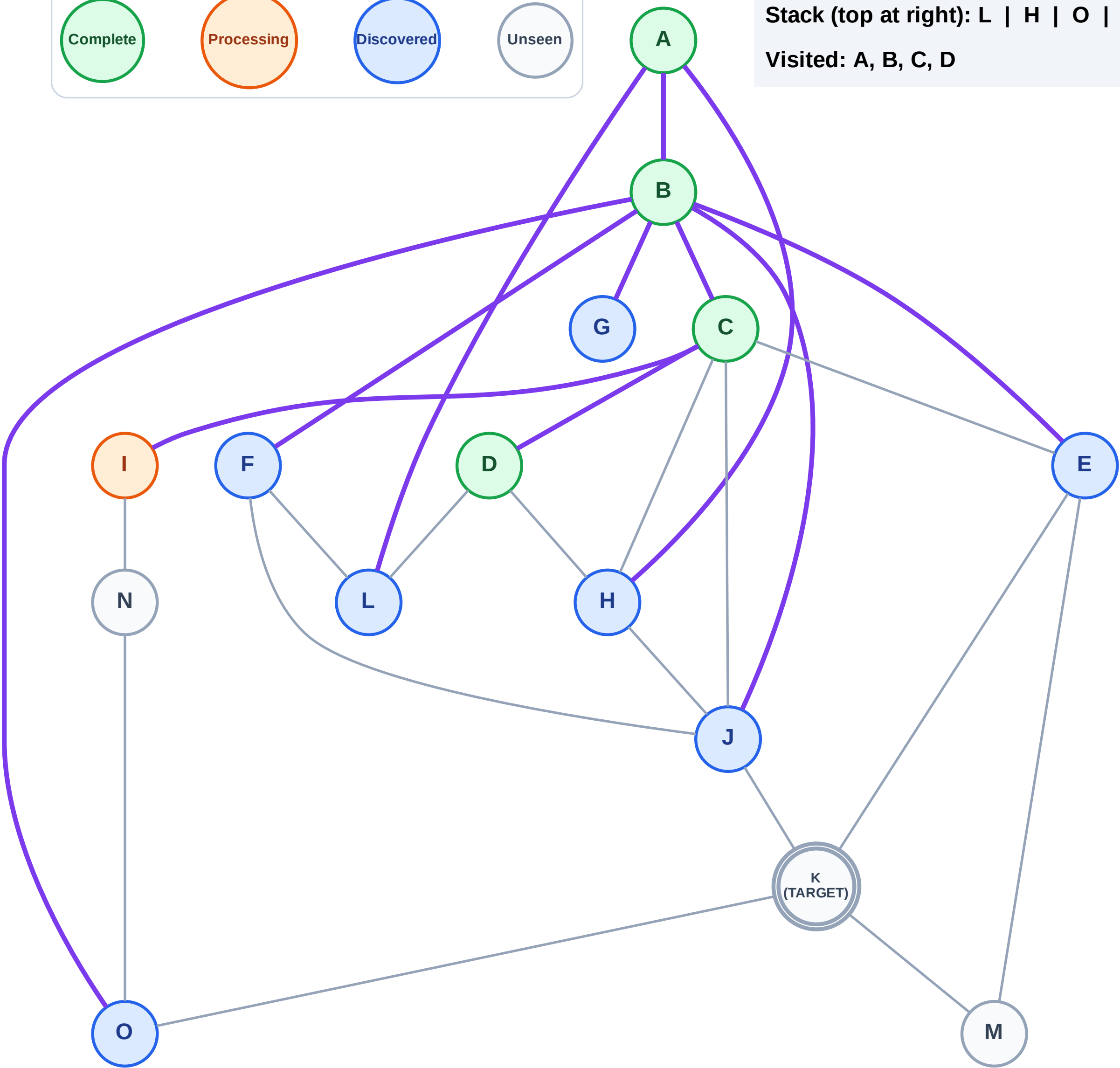
Processing

Discovered

Unseen

Stack (top at right): L | H | O | J | G | F | E

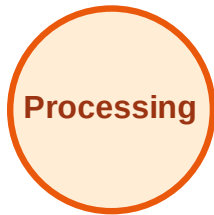
Visited: A, B, C, D



DFS Traversal

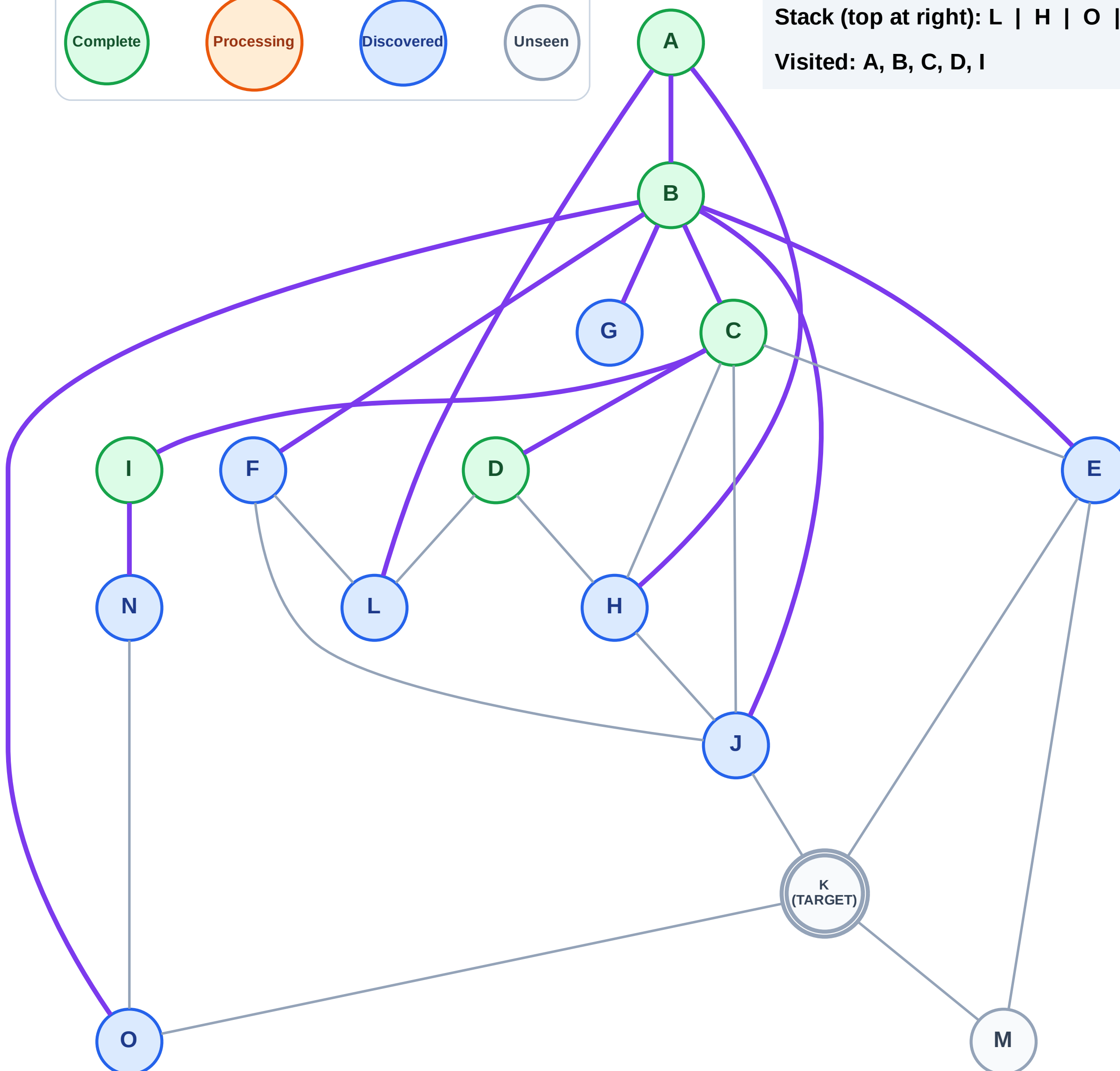
Step 11: Discover N from I

Legend



Stack (top at right): L | H | O | J | G | F | E | N

Visited: A, B, C, D, I



Step 12: Process node N

Step 12: Process node N



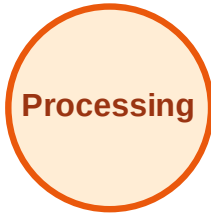
DFS Traversal

Step 13: No new neighbors from N

Legend



Complete



Processing



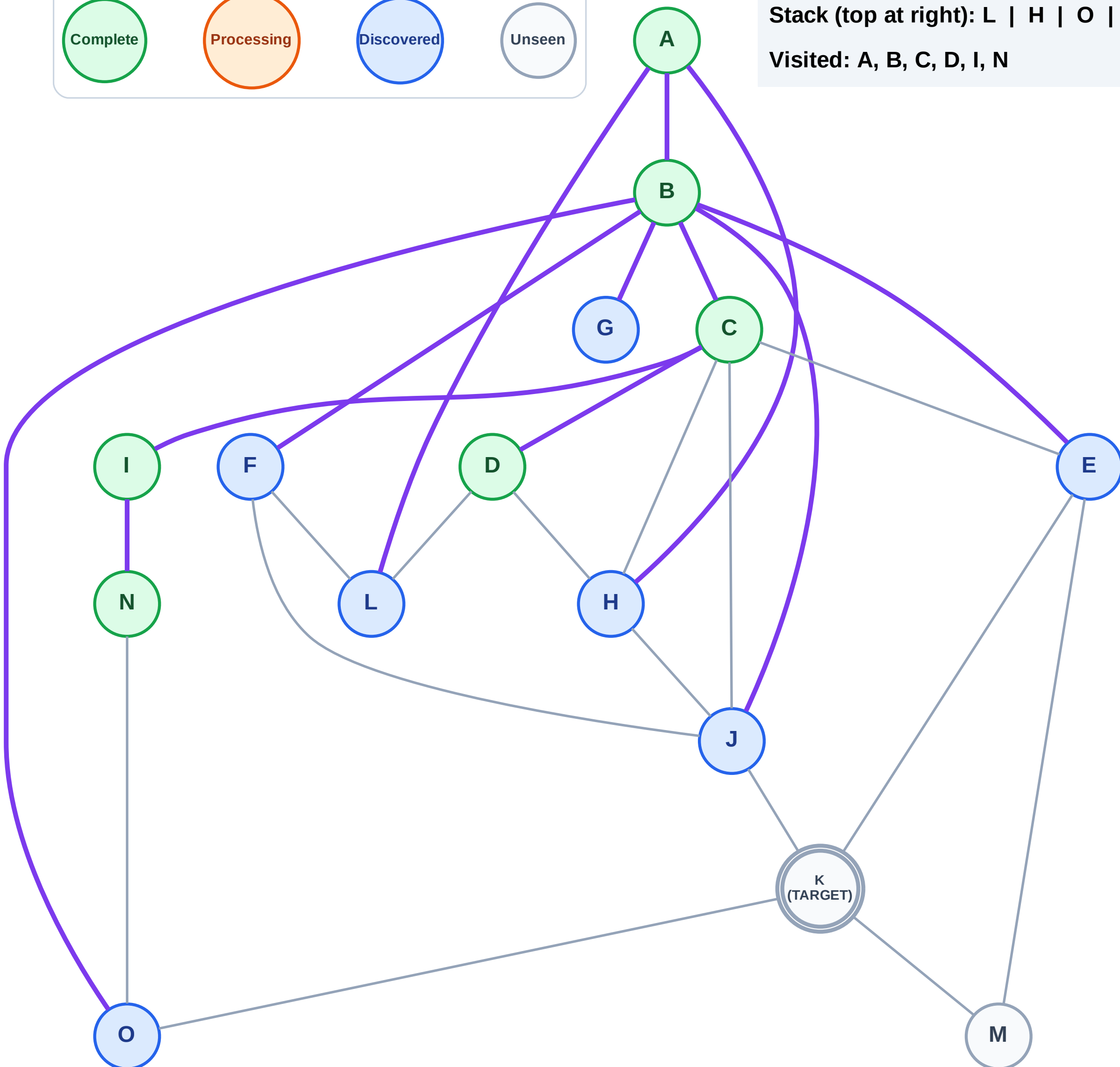
Discovered



Unseen

Stack (top at right): L | H | O | J | G | F | E

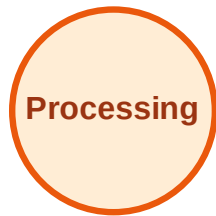
Visited: A, B, C, D, I, N



DFS Traversal

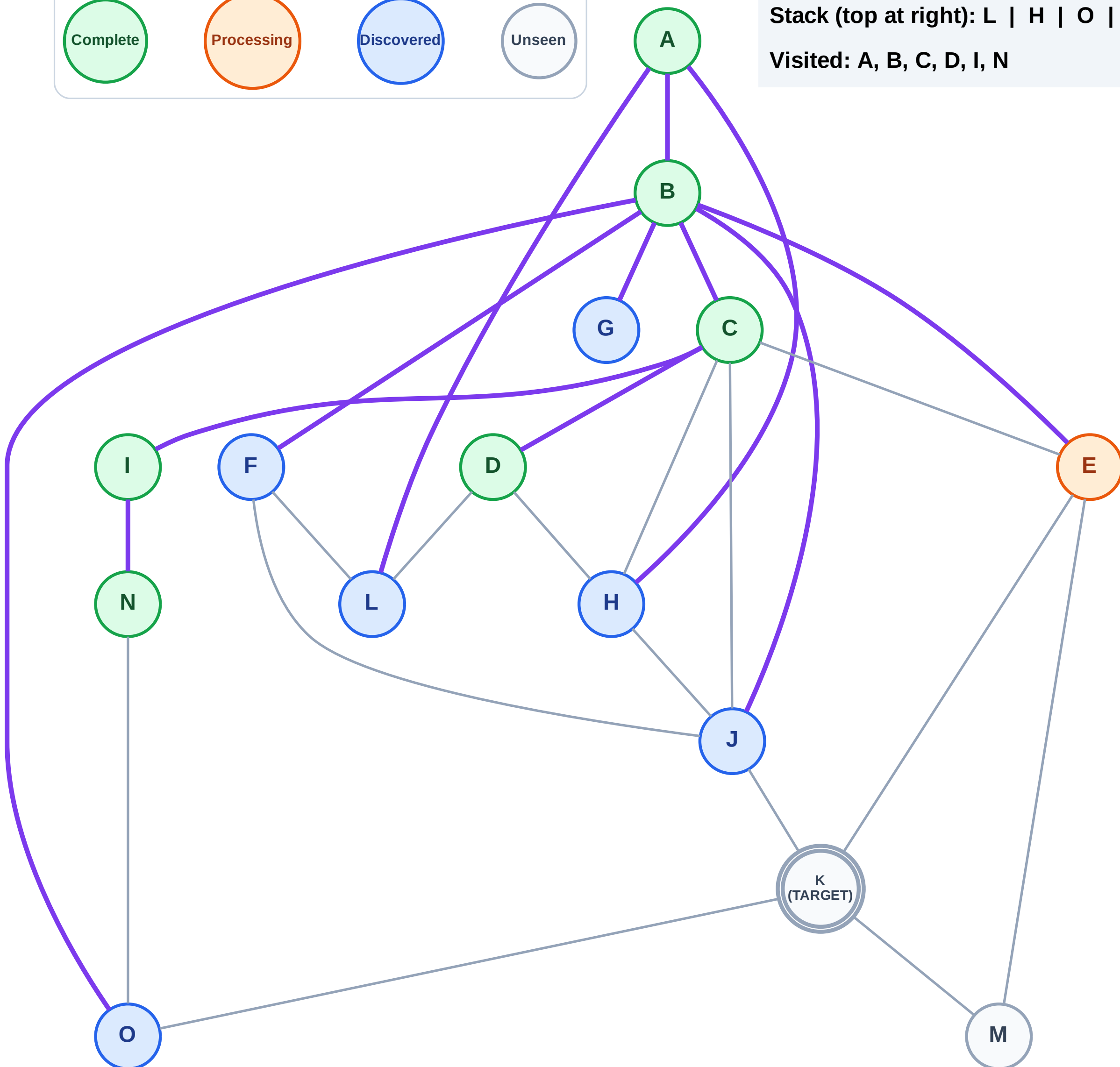
Step 14: Process node E

Legend



Stack (top at right): L | H | O | J | G | F

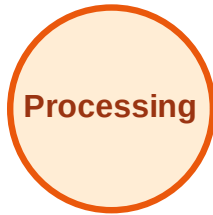
Visited: A, B, C, D, I, N



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → E → K

Legend



Stack (top at right): L | H | O | J | G | F | M

Visited: A, B, C, D, E, I, K, N

